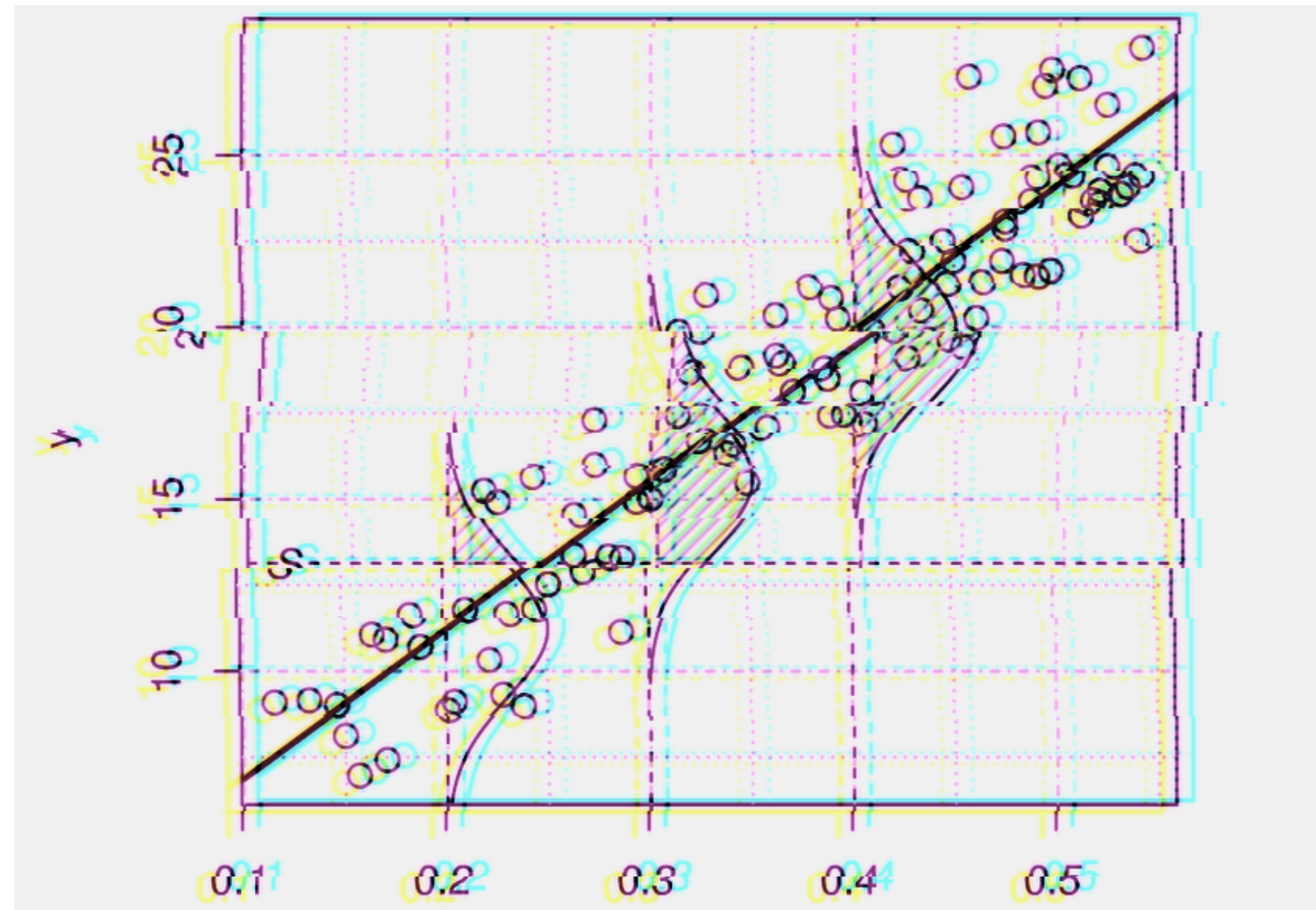
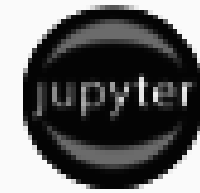


[Tuxae]



[Ateliers Data & ML] / [2. Modèles linéaires généralisés]

Introduction



Jupyter-python

The JupyterLab IDE with Python, Julia, and a collection of standard data science packages.

[Learn more](#)

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<https://shorturl.at/K7uqR>

Modèle ?

Modèle ?

Modèle ?

→
prédire

Modèle ?

<i>x1</i>	<i>x2</i>	<i>x3</i>

→
prédire

<i>y1</i>	<i>y2</i>

Modèle ?

x1	x2	x3

→
prédire

y

Modèle ?

X

$x1$	$x2$	$x3$

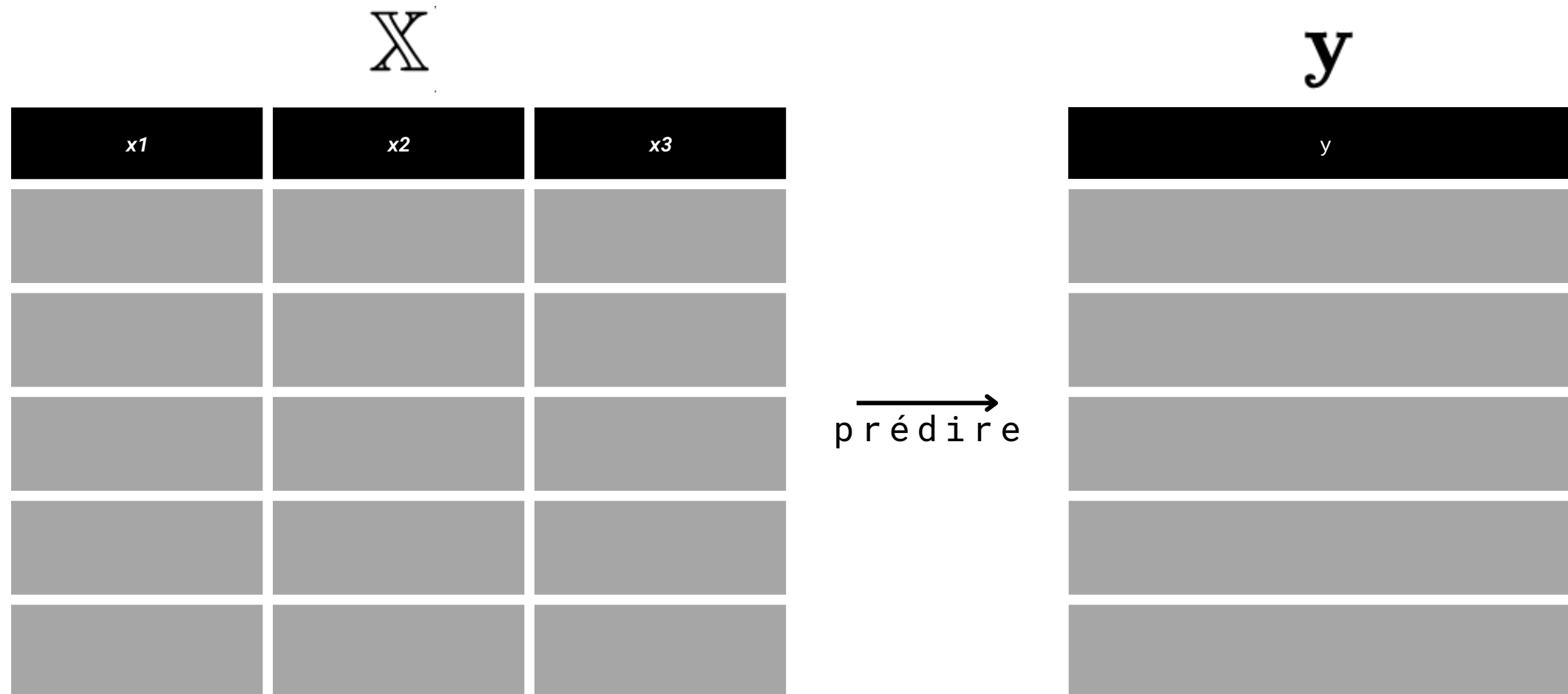
→
prédire

y

y

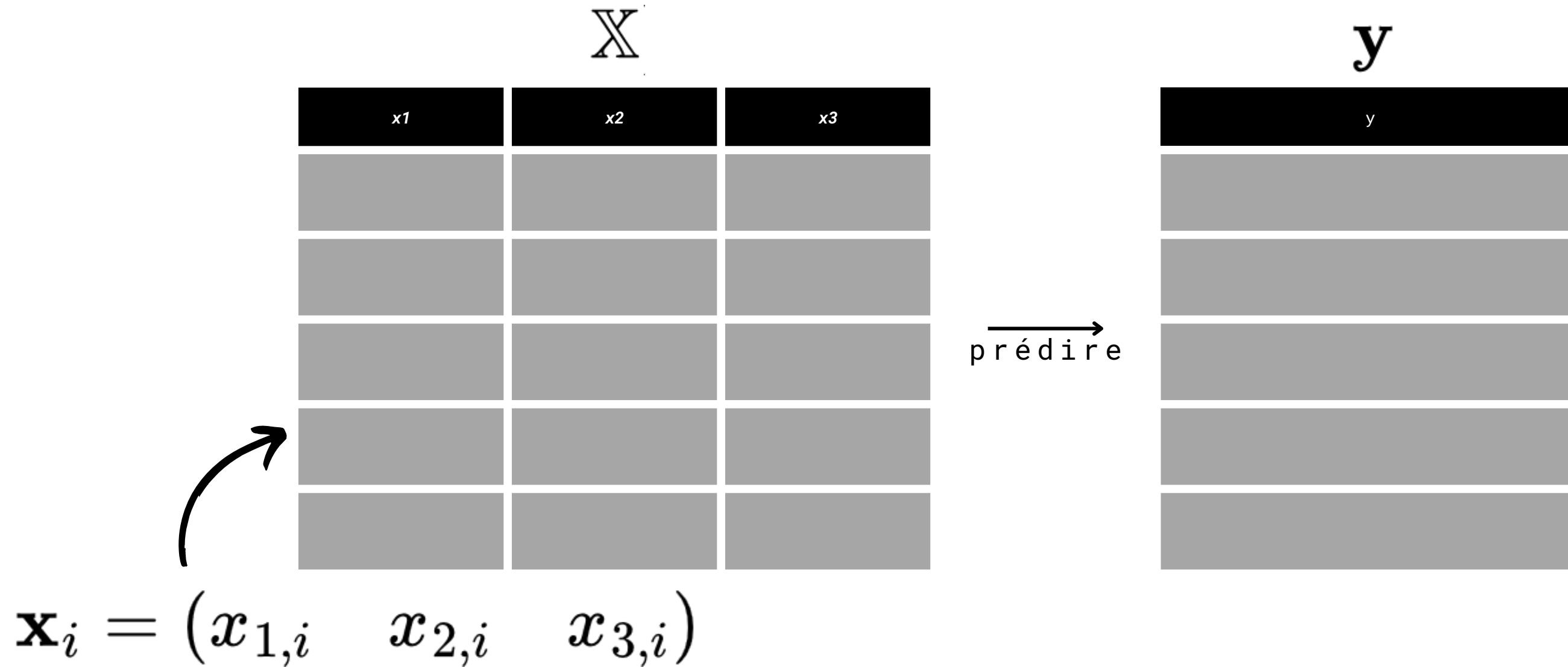
Modèle ?

$$f(\mathbb{X}) = \mathbf{y}$$



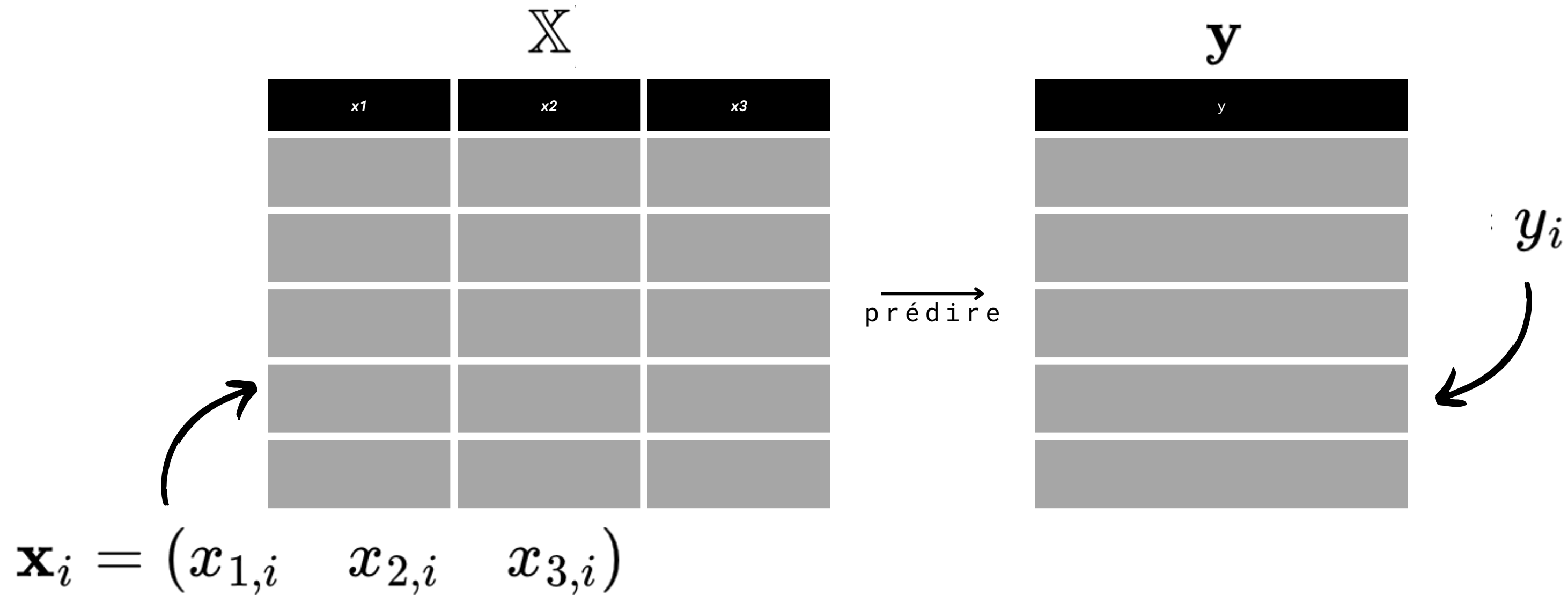
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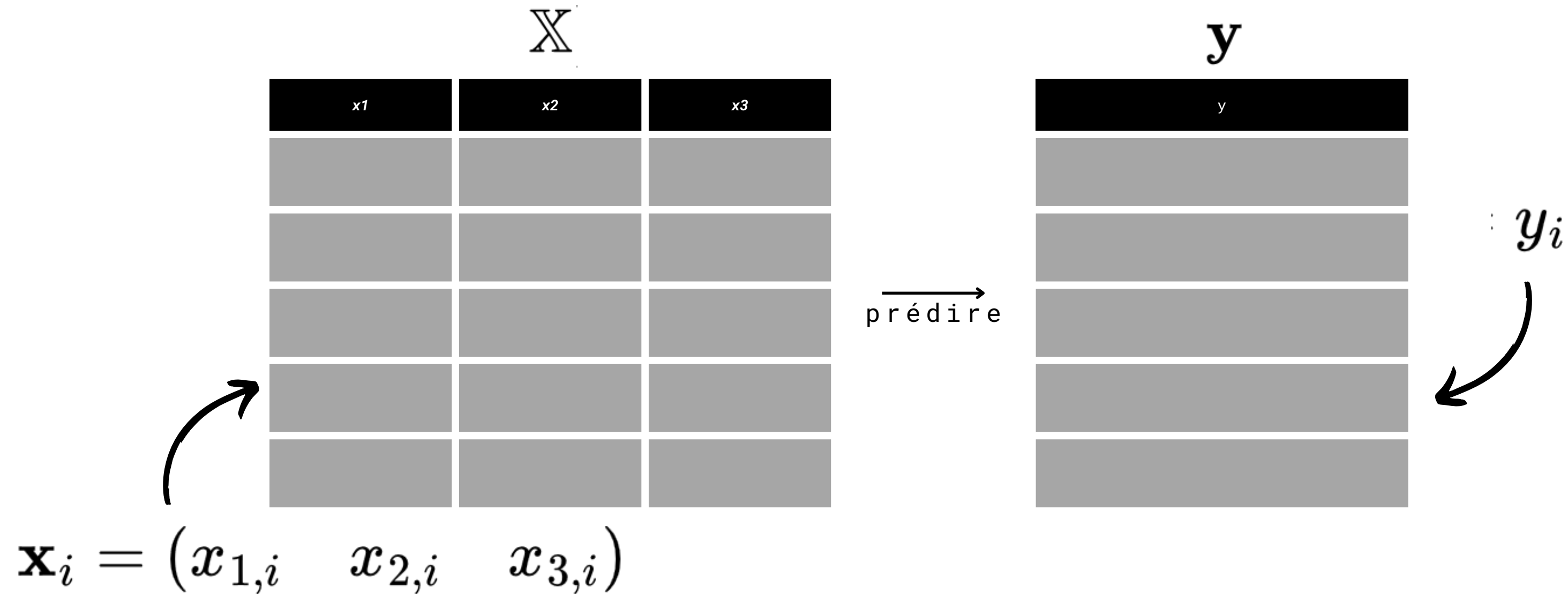
Modèle ?

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Modèle ?

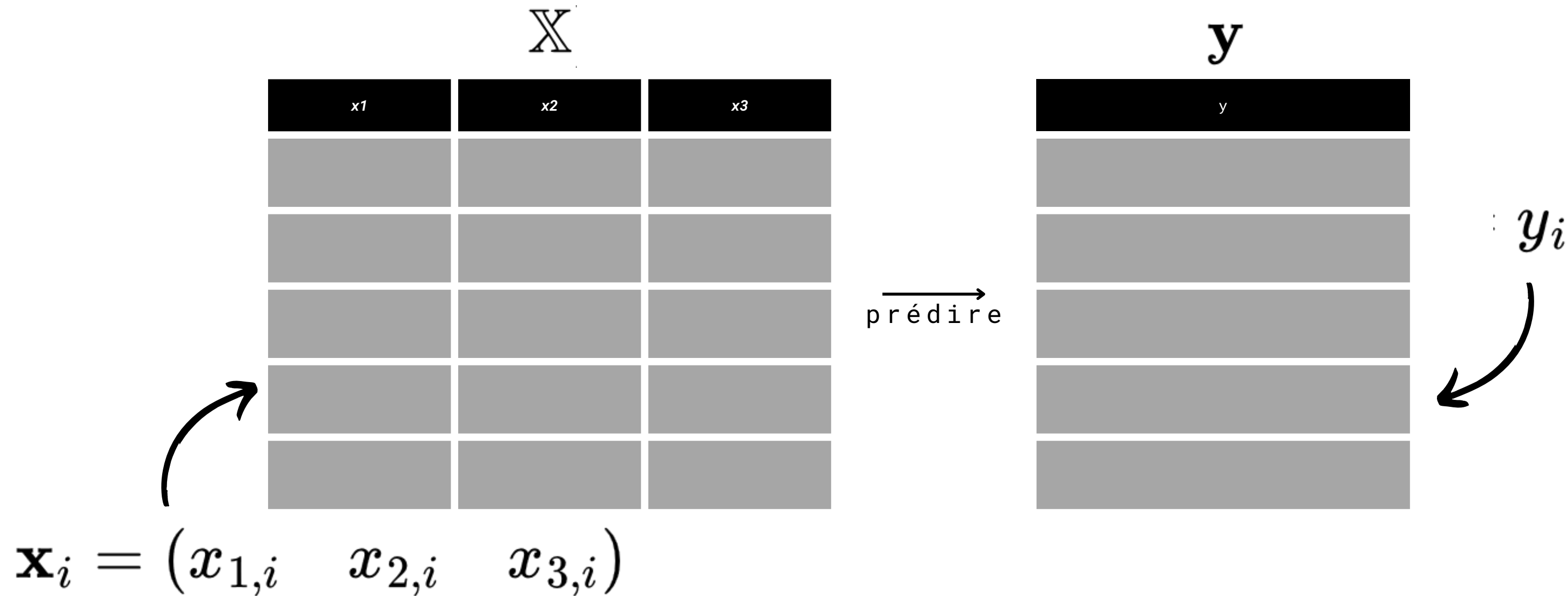
$$f(\mathbf{x}_i) = y_i \longrightarrow f(\mathbb{X}) = \mathbf{y}$$



Modèle ?

$$f(\mathbf{x}_i) = y_i \longrightarrow f(\mathbb{X}) = \mathbf{y}$$

$$h \approx f$$



Modèle ?

$$f(\mathbf{x}_i) = y_i \longrightarrow f(\mathbb{X}) = \mathbf{y} \quad \boxed{h \approx f}$$

Modèle ?

$$f(\mathbf{x}_i) = y_i \longrightarrow f(X) = \mathbf{y} \quad \boxed{h \approx f}$$

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Modèle ?

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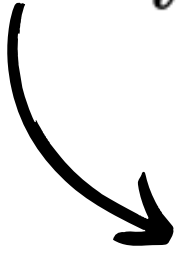
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$$\mathbf{x}_i \in \mathbb{R}^n, \quad y_i \in \mathbb{R}$$

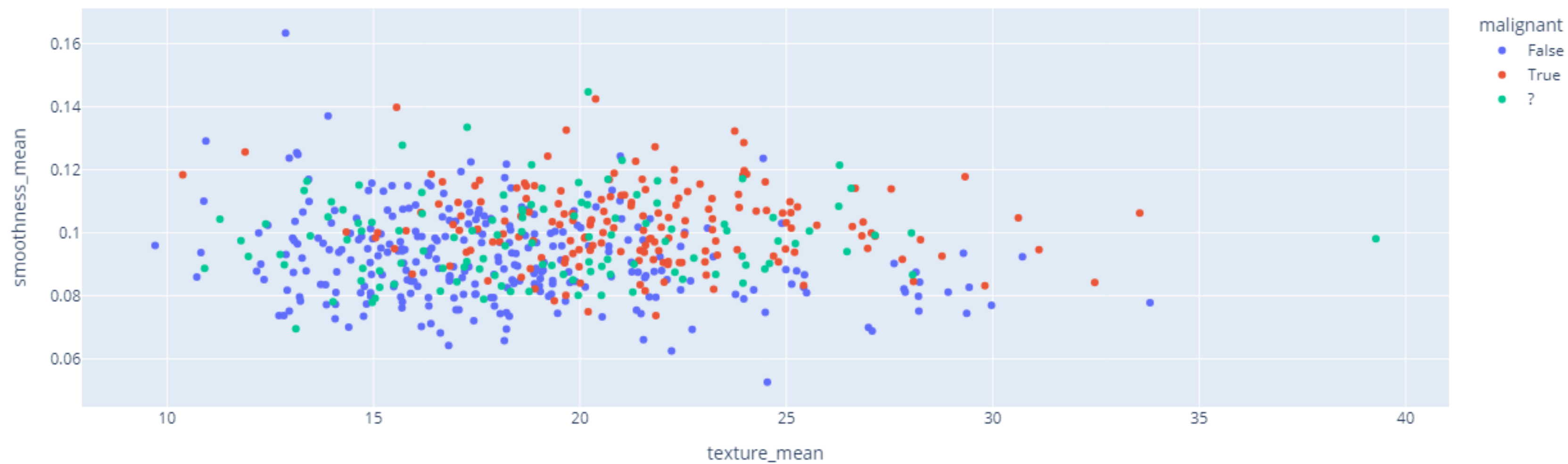
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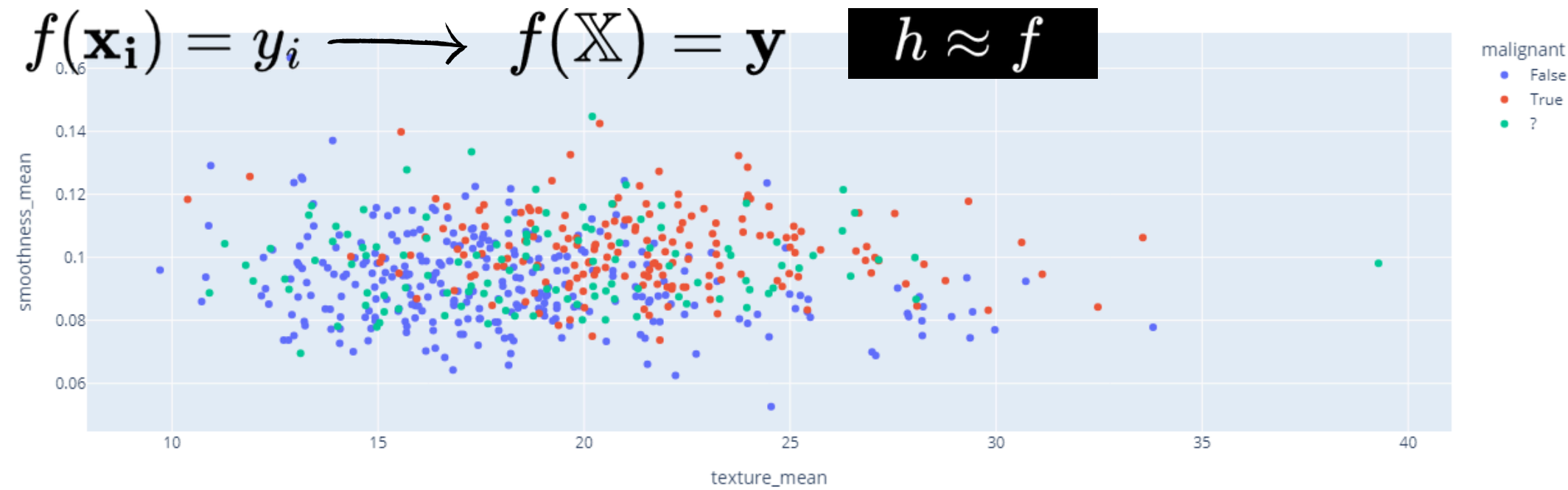
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Modèle ?



Modèle ?



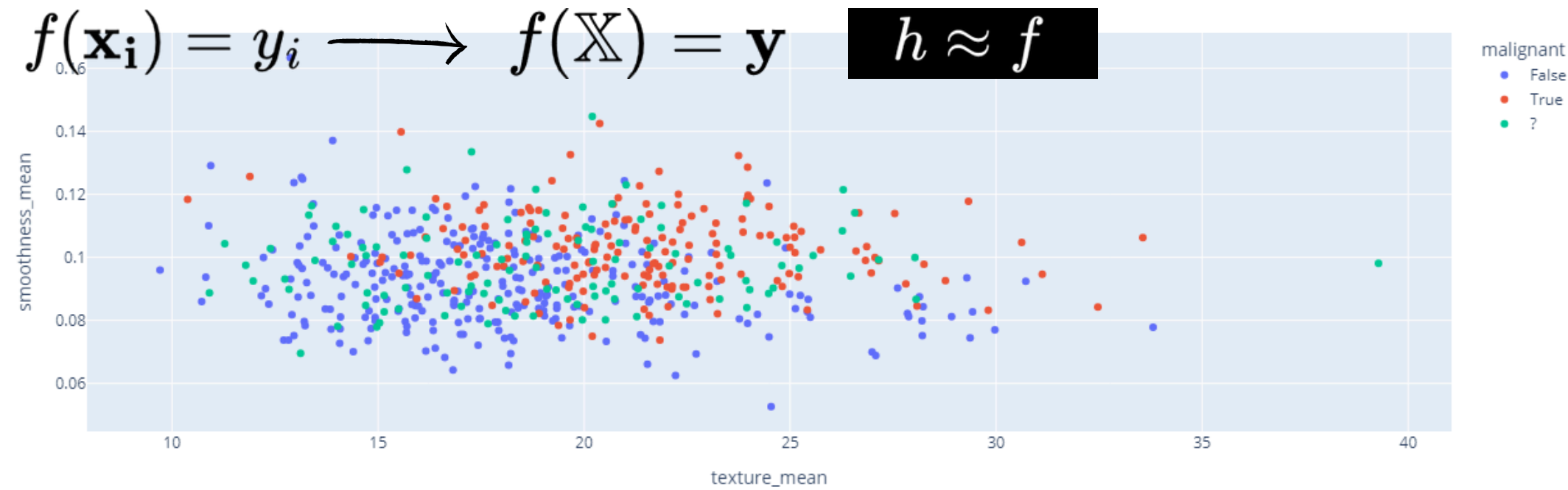
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$$\mathbf{x}_i \in \mathbb{R}^n, \quad y_i \in \mathbb{R}$$

$$y_i \approx h(\mathbf{x}_i)$$

$$h(\mathbf{x}_i) = \frac{1}{k} \sum_{j \in \mathcal{N}_k(\mathbf{x}_i)} y_j$$

Modèle ?



< HYPOTHÈSES >

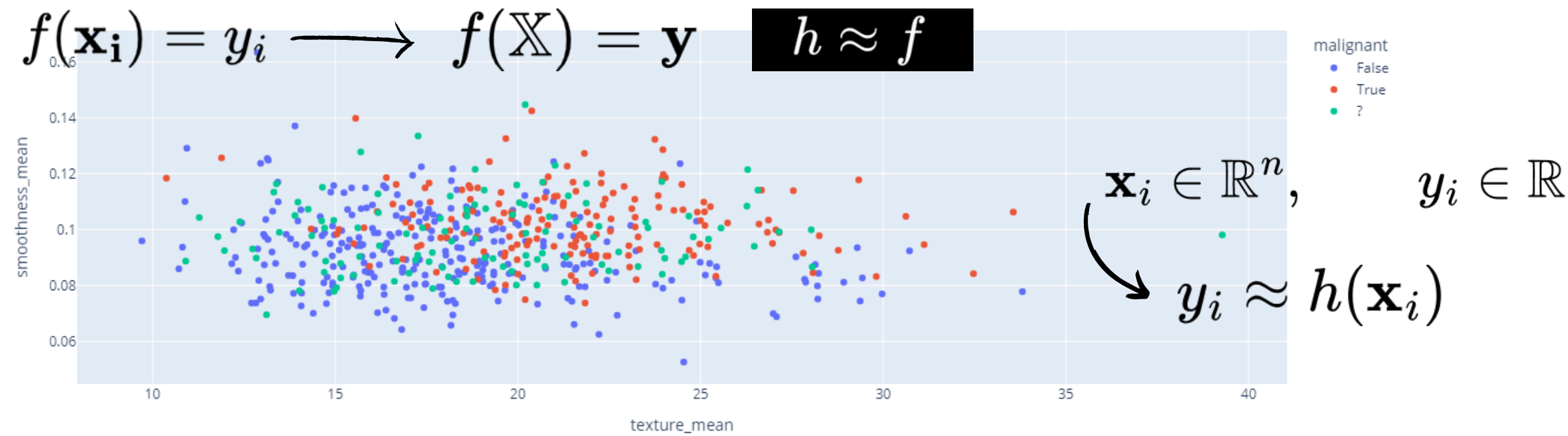
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↘

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↘

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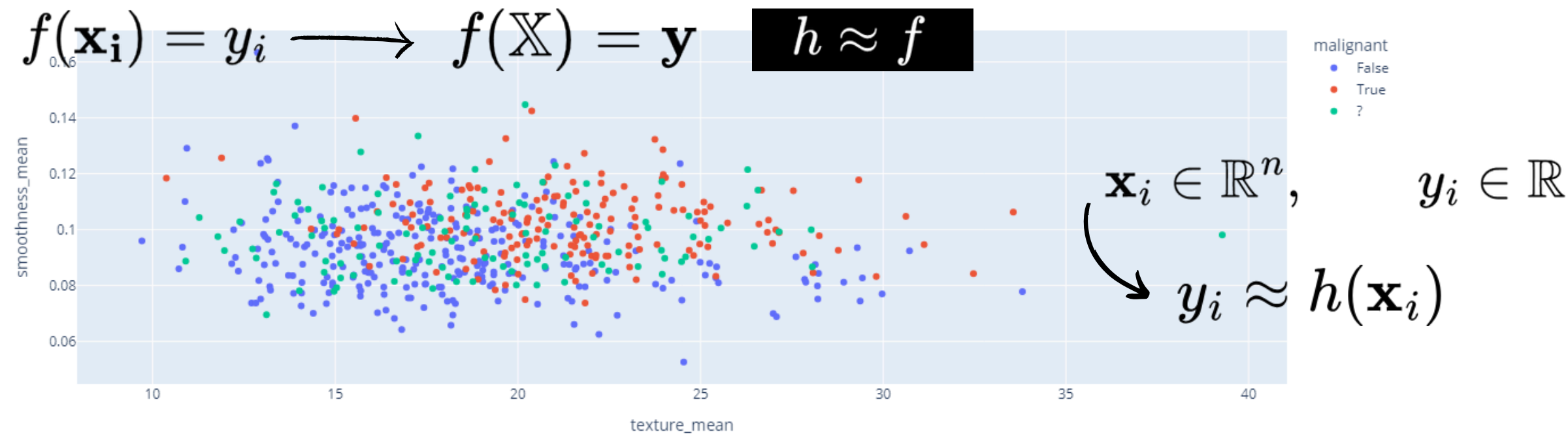


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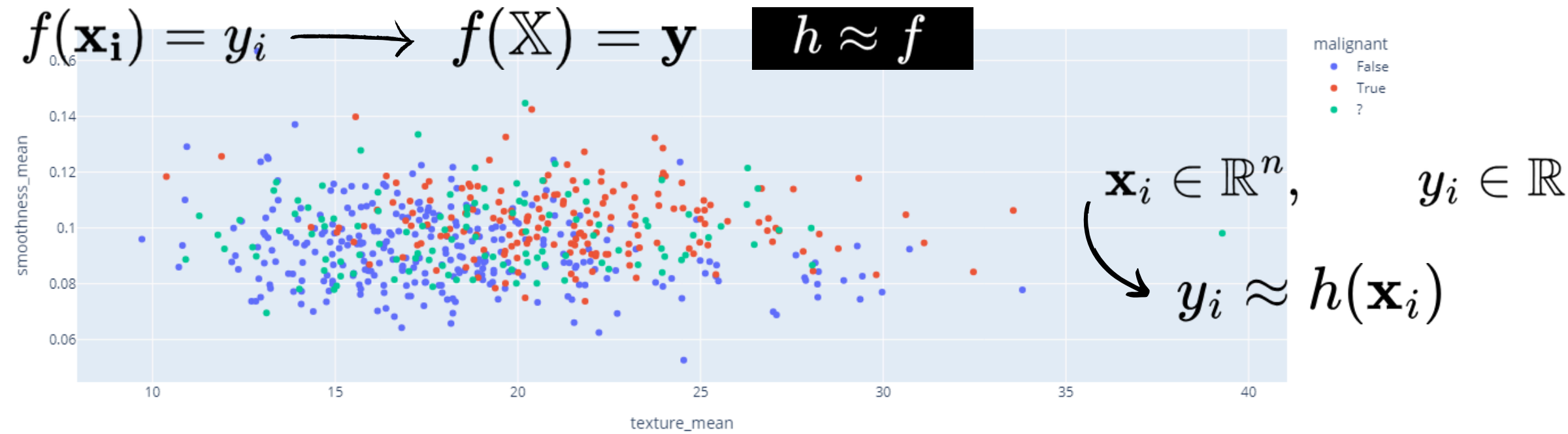
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< HYPOTHÈSES >

$$\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_i, \dots \sim \mathbf{X}$$

Modèle ?

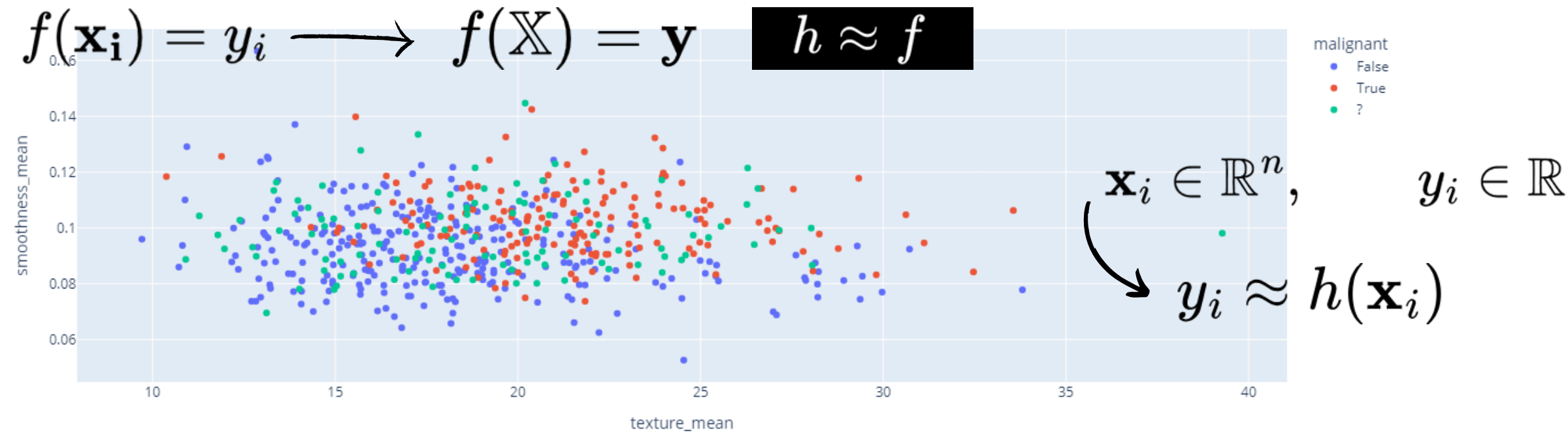


$$h(\mathbf{x}_i) = \frac{1}{k} \sum_{j \in \mathcal{N}_k(\mathbf{x}_i)} y_j \approx y_i$$

< HYPOTHÈSES >

$$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \dots \sim \mathbf{X} \quad \longrightarrow \quad \mathbf{X} = (X_1 \quad X_2 \quad X_3)$$

Modèle ?



$$h(\mathbf{x}_i) = \frac{1}{k} \sum_{j \in \mathcal{N}_k(\mathbf{x}_i)} y_j \approx y_i$$

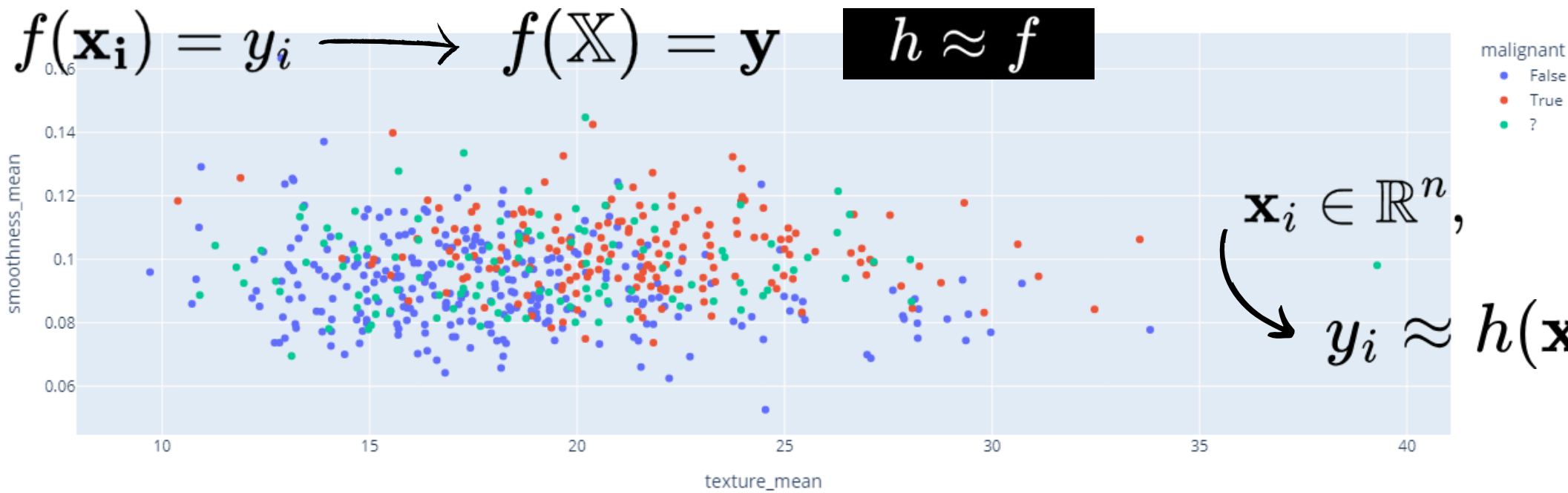
< HYPOTHÈSES >

$$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \dots \sim \mathbf{X}$$

$$y_1, y_2, \dots, y_i, \dots \sim Y$$

$$\mathbf{X} = (X_1 \quad X_2 \quad X_3)$$

Modèle ?



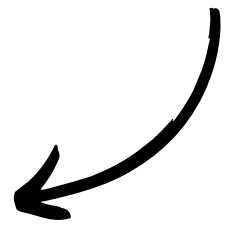
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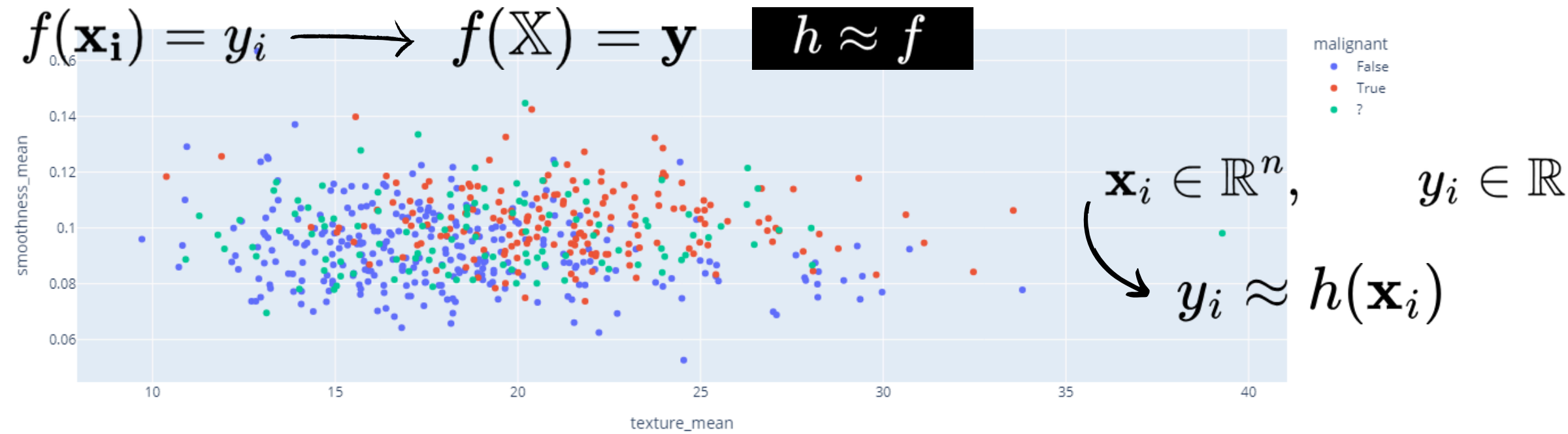
$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \dots \sim \mathbf{X}$ $\longrightarrow \mathbf{X} = (X_1 \quad X_2 \quad X_3)$

$y_1, y_2, \dots, y_i, \dots \sim Y$

$\mathbb{E}[Y | \mathbf{X}]$



Modèle ?



$$h(\mathbf{x}_i) = \frac{1}{k} \sum_{j \in \mathcal{N}_k(\mathbf{x}_i)} y_j \approx y_i$$

< HYPOTHÈSES >

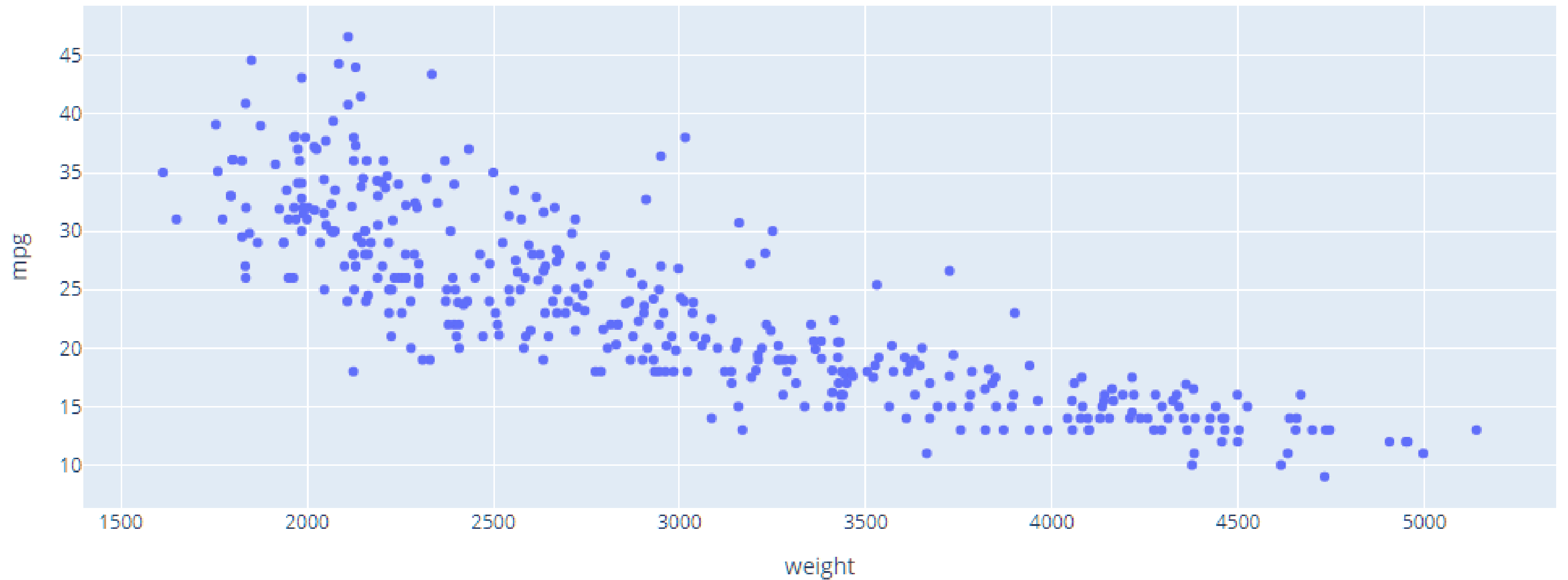
$$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \dots \sim \mathbf{X}$$

$$\mathbf{X} = (X_1 \quad X_2 \quad X_3)$$

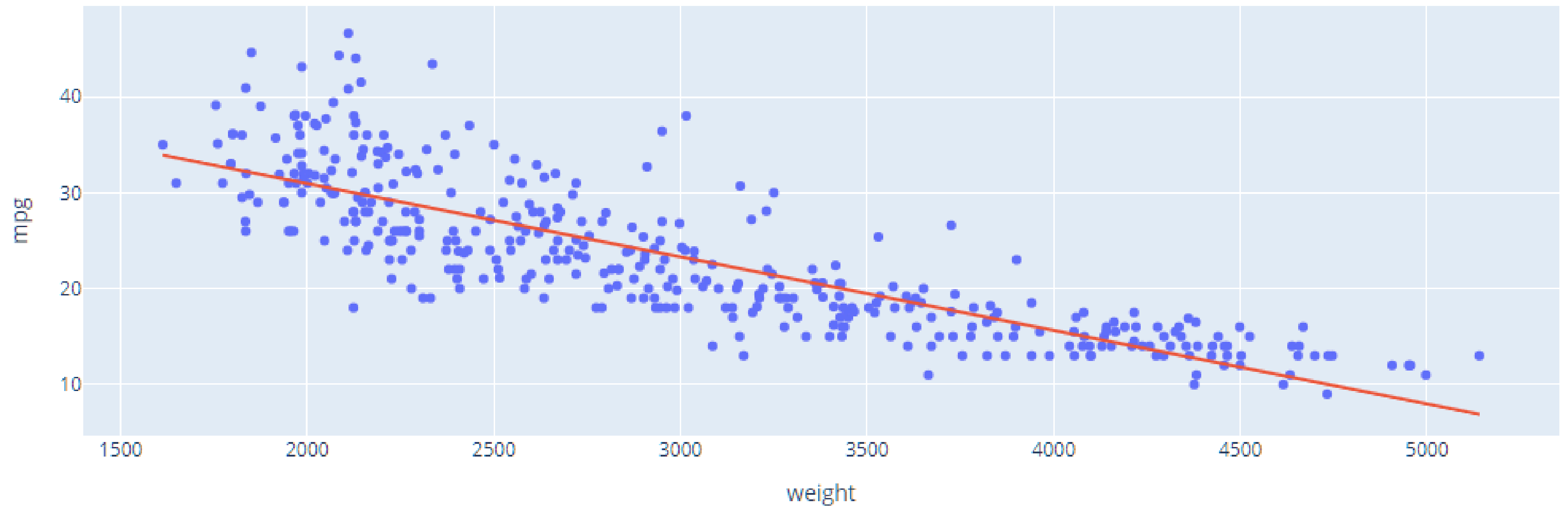
$$y_1, y_2, \dots, y_i, \dots \sim Y$$

$$\mathbb{E}[Y | \mathbf{X}] = h(\mathbf{X})$$

Régression Linéaire



Régression Linéaire



R é g r e s s i o n L i n é a i r e

$$y_i = ax_i + b + \varepsilon_i$$

R é g r e s s i o n L i n é a i r e

$$y_i = ax_i + b + \varepsilon_i \qquad h(x) = ax + b$$

R é g r e s s i o n L i n é a i r e

$$y_i = ax_i + b + \varepsilon_i \qquad h(x) = ax + b$$

$$h(\mathbf{X}) = \theta_0 + \theta_1 X_1 + \theta_2 X_2 + \cdots + \theta_n X_n$$

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$$X_0 = 1$$

R é g r e s s i o n L i n é a i r e

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R é g r e s s i o n L i n é a i r e

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$$h(\mathbf{X}) = \theta^\top \mathbf{X}$$

R é g r e s s i o n L i n é a i r e

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$$h(\mathbf{X}) = \theta^\top \mathbf{X}$$

$$Y = \theta^\top \mathbf{X} + \varepsilon$$

Régression Linéaire

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$$\mathbb{E} [\varepsilon \mid \mathbf{X}] = 0$$

Régression Linéaire

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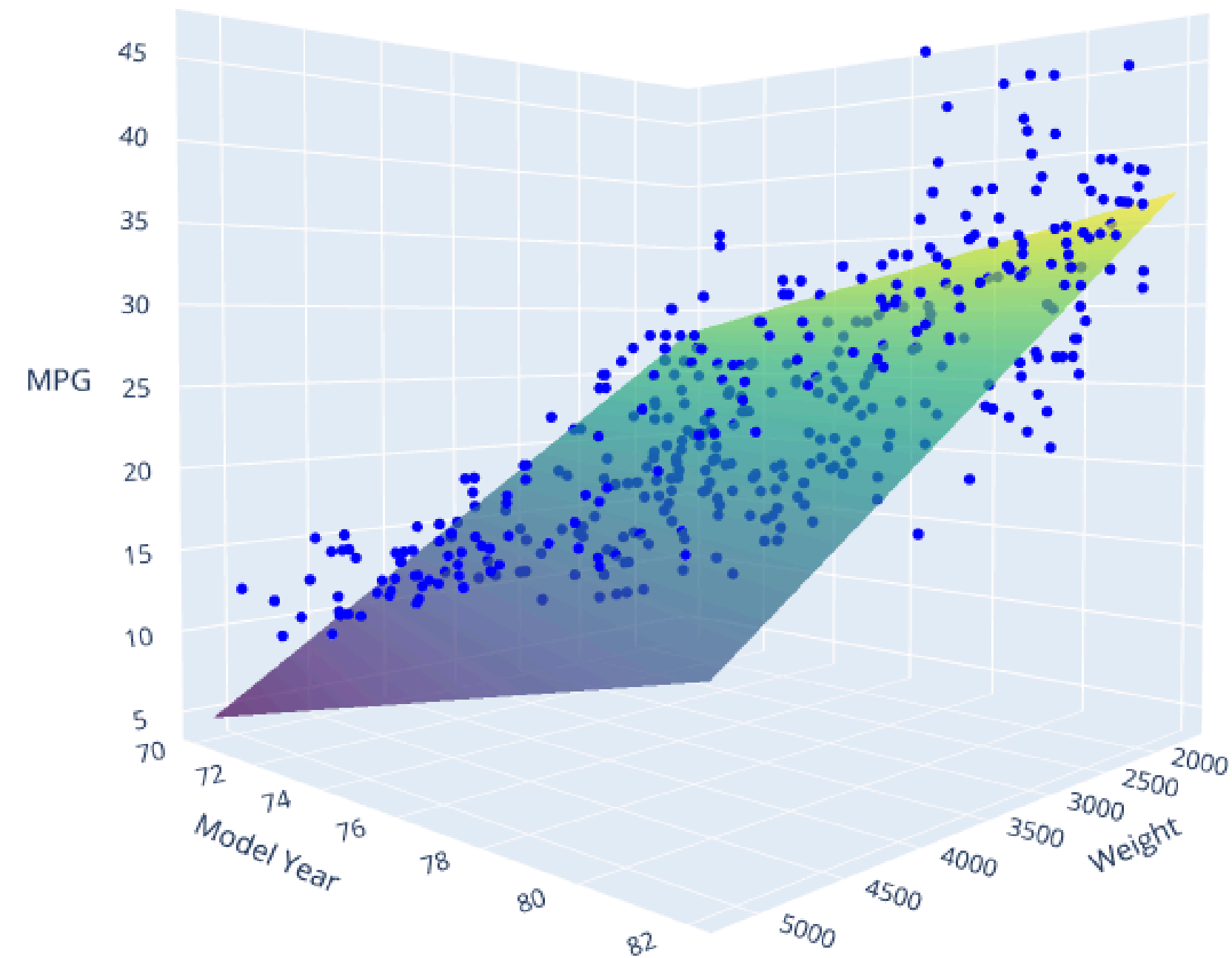
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$$h(\mathbf{X}) = \theta^\top \mathbf{X}$$

$$Y = \theta^\top \mathbf{X} + \varepsilon$$

$$\mathbb{E}[\varepsilon \mid \mathbf{X}] = 0 \implies \mathbb{E}[Y \mid \mathbf{X}] = \theta^\top \mathbf{X}$$

Régression Linéaire



Régression Linéaire

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$$\mathbb{E}[\varepsilon \mid \mathbf{X}] = 0 \implies \mathbb{E}[Y \mid \mathbf{X}] = \theta^\top \mathbf{X}$$

E s t i m a t e u r

$$e(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p) \approx \theta$$

E s t i m a t e u r

$$\underbrace{e(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p)}_{\widehat{\theta}_p} \approx \theta$$

E s t i m a t e u r

$$\underbrace{e(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p)}_{\hat{\theta}_p} \approx \theta$$

$$\hat{\theta}_p \xrightarrow{p \rightarrow +\infty} \theta$$

E s t i m a t e u r

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Estimateur

$$\underbrace{e(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p)}_{\hat{\theta}_p} \approx \theta$$

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$$\hat{\theta} = \operatorname{argmin} \left[\operatorname{loss} \left(y, y_{\text{pred}} \{ \mathbf{x}_i, y_i \}_{1 \leq i \leq p} \right) \right]$$

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E s t i m a t e u r

$$y_i = \hat{\theta}^\top \mathbf{x}_i + \varepsilon_i$$

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E s t i m a t e u r

$$y_i = \hat{\theta}^\top \mathbf{x}_i + \varepsilon_i \iff \varepsilon_i = y_i - \hat{\theta}^\top \mathbf{x}_i$$

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E s t i m a t e u r

$$y_i = \hat{\theta}^\top \mathbf{x}_i + \varepsilon_i \iff \varepsilon_i = y_i - \hat{\theta}^\top \mathbf{x}_i$$

$$\text{MSE} = \frac{1}{p} \sum_{i=1}^p (y_i - \mathbf{y_pred}_i)^2$$

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E s t i m a t e u r O L S

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$$\hat{\theta}_{OLS} = \operatorname{argmin}_{\theta} [\text{MSE}]$$

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E s t i m a t e u r O L S

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Estimateur OLS

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$$\hat{\theta}_{OLS} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

$$\hat{\theta} = \operatorname{argmin} \left[\operatorname{loss} \left(\mathbf{y}, \mathbf{y_pred} \{ \mathbf{x}_i, y_i \}_{1 \leq i \leq p} \right) \right]$$

E s t i m a t e u r O L S

$$\text{MSE} = \frac{1}{p} \sum_{i=1}^p (y_i - \mathbf{y_pred}_i)^2$$

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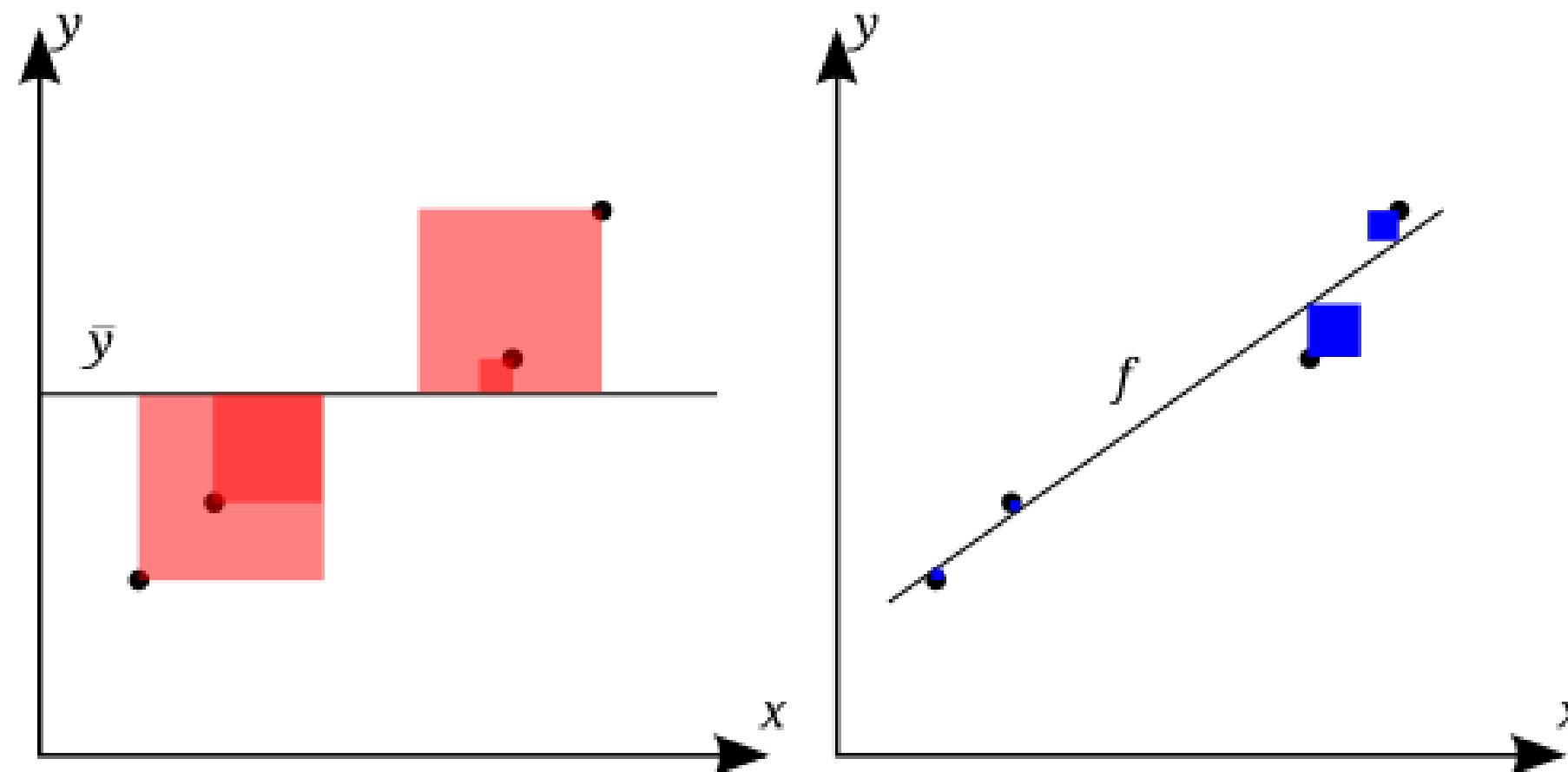
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$$\hat{\theta}_{OLS} = \mathbf{X}^+ \mathbf{y}$$

Coefficient de détermination

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} = \text{corr}(y, y_{\text{pred}})^2$$



$$\hat{\theta}_{OLS} = \operatorname{argmin}_{\theta} \left[\frac{1}{p} \sum_{i=1}^p (y_i - \theta^{\top} \mathbf{x}_i)^2 \right]$$

L a s s o

L a s s o

$$\hat{\theta}_{OLS} = \operatorname{argmin}_{\theta} \left[\frac{1}{p} \sum_{i=1}^p (y_i - \theta^{\top} \mathbf{x}_i)^2 \right]$$

$$\hat{\theta}_{\text{Lasso}} = \operatorname{argmin}_{\theta} \left[\frac{1}{p} \sum_{i=1}^p (y_i - \theta^{\top} \mathbf{x}_i)^2 + \alpha \|\theta\|_1 \right]$$

R i d g e

$$\hat{\theta}_{OLS} = \operatorname{argmin}_{\theta} \left[\frac{1}{p} \sum_{i=1}^p (y_i - \theta^{\top} \mathbf{x}_i)^2 \right]$$

$$\hat{\theta}_{\text{Lasso}} = \operatorname{argmin}_{\theta} \left[\frac{1}{p} \sum_{i=1}^p (y_i - \theta^{\top} \mathbf{x}_i)^2 + \alpha \|\theta\|_1 \right]$$

$$\hat{\theta}_{\text{Ridge}} = \operatorname{argmin}_{\theta} \left[\frac{1}{p} \sum_{i=1}^p (y_i - \theta^{\top} \mathbf{x}_i)^2 + \alpha \|\theta\|_2^2 \right]$$

I n t u i t i o n

Elastic-Net

$$\hat{\theta}_{\text{EN}} = \arg \min_{\theta} \left[\frac{1}{2p} \sum_{i=1}^p (y_i - \theta^{\top} \mathbf{x}_i)^2 + \alpha \|\theta\|_1 + \frac{\alpha(1 - \rho)}{2} \|\theta\|_2^2 \right]$$

Elastic-Net

$$\hat{\theta}_{\text{EN}} = \arg \min_{\theta} \left[\frac{1}{2p} \sum_{i=1}^p (y_i - \theta^{\top} \mathbf{x}_i)^2 + \alpha \|\theta\|_1 + \frac{\alpha(1 - \rho)}{2} \|\theta\|_2^2 \right]$$

R é g r e s s i o n l o g i s t i q u e

$$\sigma(z) = \frac{1}{1 + e^{-z}} \qquad \log \sigma(z) = -\log(1 + e^{-z})$$

$$\mathbb{E}[Y \mid \mathbf{X}] = \sigma(\theta^\top \mathbf{X})$$

Régression logistique

R é g r e s s i o n l o g i s t i q u e

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$$\hat{\theta}_{\text{LogReg}} = \operatorname{argmin}_{\theta} \left[\frac{1}{p} \sum_{i=1}^p \log(1 + e^{-y_i \theta^\top x_i}) \right]$$

[Et voilà . . .]
Fin de l'atelier :)

SIAHAAN -- GENSOLLEN Rémy, 2025